Available online at www.sciencedirect.com
SCIENCE DIRECT®
ELSEVIER Applied Mathematics and Computation 167 (2005) 94–107
APPLIED MATHEMATICS AND COMPUTATION
www.elsevier.com/locate/amc

Convergence of descent method without line search

Zhen-Jun Shi [a,b,*], Jie Shen [c]

[a] College of Operations Research and Management, Qufu Normal University (Rizhao College), Rizhao, Shandong 276826, PR China
[b] Institute of Computational Mathematics and Scientific/Engineering Computing, Chinese Academy of Sciences, P.O. Box 2719, Beijing 100080, PR China
[c] Department of Computer and Information Science, University of Michigan, Dearborn, MI 48128, USA

Abstract

Line search method is sometimes a useful and efficient technique for solving unconstrained optimization problems, especially for solving small and middle scale problems. However, A line search procedure is necessary at each iteration, leading to a significant amount of computation. In order to reduce the evaluations of objective functions and gradients, the line search procedure should be avoided in algorithm design. In this paper we propose a new descent method without line search for unconstrained optimization problems. The algorithm is required to estimate some parameters at each iteration. We analyze theoretically the global convergence of the new algorithm under mild conditions. These theoretical conclusions can help us in designing new efficient methods for optimization problems.
© 2004 Elsevier Inc. All rights reserved.

Keywords: Unconstrained optimization; Descent method; Line search; Convergence

* Corresponding author. Address: College of Operations Research and Management, Qufu Normal University (Rizhao College), Rizhao, Shandong 276826, PR China.
E-mail addresses: zjshi@qrnu.edu.cn (Z.-J. Shi), shen@umich.edu (J. Shen).

0096-3003/$ - see front matter © 2004 Elsevier Inc. All rights reserved.
doi:10.1016/j.amc.2004.06.097

1. Introduction

Line search methods can be used for solving an unconstrained minimization problem

$$\min f(x), \quad x \in R^n, \tag{1}$$

where $f: R^n \to R^1$ is a continuously differentiable function and R^n denotes the n-dimensional Euclidean space, which take the form

$$x_{k+1} = x_k + \alpha_k d_k, \tag{2}$$

in which d_k is a descent direction and α_k a positive step-size. Letting x_k be the current iterative point ($k = 0, 1, \ldots$), we denote $\nabla f(x_k)$ by g_k, $f(x_k)$ by f_k and $f(x^*)$ by f^*, respectively.

Many traditional descent methods for solving (1) are line search methods, such as steepest descent methods, Newton-type methods, variable metric methods (e.g., [12]), and conjugate gradient methods

$$d_k = \begin{cases} -g_k, & \text{if } k = 1, \\ -g_k + \beta_k d_{k-1}, & \text{if } k \geqslant 2, \end{cases} \tag{3}$$

where β_k is a parameter (e.g., [4,8,9,17], etc.), etc. In fact, a huge amount of literature is devoted to conjugate gradient methods. We cannot list all excellent ones here.

Choosing search direction d_k and determining step-size α_k along the direction at each iteration are the main task in line search methods [18]. The search direction d_k is generally required to satisfy

$$g_k^{\mathrm{T}} d_k < 0, \tag{4}$$

which guarantees that d_k is a descent direction of $f(x)$ at x_k [1,12]. In order to guarantee the global convergence, we sometimes require that d_k satisfies the sufficient descent condition

$$g_k^{\mathrm{T}} d_k \leqslant -c\|g_k\|^2, \tag{5}$$

where $c > 0$ is a constant. Moreover, the angle property is

$$\cos\langle -g_k, d_k \rangle = -\frac{g_k^{\mathrm{T}} d_k}{\|g_k\| \cdot \|d_k\|} \geqslant \eta_0, \tag{6}$$

where $\eta_0 > 0$ is a constant.

In line search methods we need to minimize $f(x)$ over a ray $S_k = \{x_k + \alpha d_k | \alpha > 0\}$ at each iteration.

Generally, there are many line search rules to choose α_k along the ray $S_k = \{x_k + \alpha d_k | \alpha > 0\}$ (e.g., [10,11,16], etc). The commonly-used line search rules are as follows.

(a) *Minimization rule*: At each iteration, α_k is selected so that

$$f(x_k + \alpha_k d_k) = \min_{\alpha>0} f(x_k + \alpha d_k). \tag{7}$$

(b) *Approximate minimization rule*: At each iteration, α_k is selected so that

$$\alpha_k = \min\{\alpha | g(x_k + \alpha d_k)^{\mathrm{T}} d_k = 0, \alpha > 0\}. \tag{8}$$

(c) *Armijo rule*: Let scalars $s_k = -\frac{g_k^{\mathrm{T}} d_k}{L\|d_k\|^2}$, $L > 0$, $\beta \in (0,1)$, and $\sigma \in (0, \frac{1}{2})$. Set $\alpha_k = \beta^{m_k} s_k$, where m_k is the first nonnegative integer m for which

$$f_k - f(x_k + \beta^m s_k d_k) \geqslant -\sigma \beta^m s_k g_k^{\mathrm{T}} d_k, \tag{9}$$

i.e., $m = 0, 1, \ldots$ are tried successively until the inequality above is satisfied for $m = m_k$.

(d) *Limited minimization rule*: Set $s_k = -\frac{g_k^{\mathrm{T}} d_k}{L\|d_k\|^2}$, and α_k is defined by

$$f(x_k + \alpha_k d_k) = \min_{\alpha \in [0, s_k]} f(x_k + \alpha d_k), \tag{10}$$

where $L > 0$ is a constant.

(e) *Goldstein rule*: A fixed scalar $\sigma \in (0, \frac{1}{2})$ is selected, and α_k is chosen to satisfy

$$\sigma \leqslant \frac{f(x_k + \alpha_k d_k) - f_k}{\alpha_k g_k^{\mathrm{T}} d_k} \leqslant 1 - \sigma. \tag{11}$$

It is possible to show that if f is bounded below, there exists an interval of step-sizes α_k for which the relation above is satisfied, and there are fairly simple algorithms for finding such a step-size through a finite number of arithmetic operations.

(f) *Strong Wolfe rule*: α_k is chosen to satisfy simultaneously

$$f_k - f(x_k + \alpha_k d_k) \geqslant -\sigma \alpha_k g_k^{\mathrm{T}} d_k \tag{12}$$

and

$$|g(x_k + \alpha_k d_k)^{\mathrm{T}} d_k| \leqslant -\beta g_k^{\mathrm{T}} d_k, \tag{13}$$

where σ and β are some scalars with $\sigma \in (0, \frac{1}{2})$ and $\beta \in (\sigma, 1)$.

(g) *Wolfe rule*: α_k is chosen to satisfy (12) and

$$g(x_k + \alpha_k d_k)^{\mathrm{T}} d_k \geqslant \beta g_k^{\mathrm{T}} d_k. \tag{14}$$

It is obvious that any line search rule is a procedure for finding α_k. This certainly adds the number of evaluations for objective functions and gradients. Dixon [6] proposed a conjugate direction method without line search. Recently Sun and Zhang [15], Chen and Sun [5] have investigated some conjugate

gradient methods without line search. This enables us to design a general descent method without line search. Vrahatis et al. [18] presented a minimization algorithm with adaptive step-size. Shi [16] analyzed the convergence of line search methods with the above line search rules. This also helps us in finding some descent methods without line search.

In this paper we propose a new descent method without line search for unconstrained optimization problems. The algorithm is required to estimate some parameters at each iteration. We analyze theoretically the global convergence of the new algorithm under mild conditions. These theoretical conclusions can facilitate the design of new efficient methods for optimization problems.

The rest of this paper is organized as follows. In Section 2 we describe a descent method without line search. In Sections 3 and 4 we analyze its convergence and convergence rate, respectively. In Section 5 we analyze some techniques for choosing parameters used in the algorithm.

2. Decent methods without line search

We first assume that

(H_1): The function f has a lower bound on the level set $L_0 = \{x \in R^n | f(x) \leqslant f(x_0)\}$, where x_0 is given.
(H_2): The gradient $g(x)$ of $f(x)$ is uniformly continuous in an open convex set B that contains L_0.

We sometimes further assume the following conditions.

(H_3): The gradient $g(x)$ of $f(x)$ is Lipschitz continuous in an open convex set B that contains the level set L_0, i.e., there exists L such that

$$\|g(x) - g(y)\| \leqslant L\|x - y\|, \quad \forall x, y \in B.$$

Obviously, (H_3) implies (H_2).

If L in Assumption (H_3) is easy to estimate, then we can design some descent algorithm without line search.

Algorithm model (A). $x_1 \in R^n$, $\frac{1}{2} < \rho \leqslant 1$, $k := 1$;

Step 1: If $\|g_k\| = 0$, then stop! else goto Step 2;
Step 2: Find a descent direction d_k that satisfies (4) and an integer $m_k > 0$;
Step 3: $x_{k+1} = x_k + \alpha_k d_k$, where

$$\alpha_k = -\frac{g_k^{\mathrm{T}} d_k}{L_k \|d_k\|^2}, \tag{15}$$

in which $m_k L \geqslant L_k \geqslant \rho L$;

Step 4: $k := k + 1$, goto Step 1.

(H_4): $f(x)$ is twice continuously differentiable in an open convex set B that contains the level set L_0 and its Hessian matrix $H(x) = \nabla^2 f(x)$ satisfies

$$\|\nabla^2 f(x)\| \leqslant M, \quad \forall x \in B,$$

where $M > 0$ is a constant.

It is easy to prove that (H_4) implies (H_3). If M is easy to evaluate, then we obtain the following descent method without line search.

Algorithm model (B). $x_1 \in R^n$, $\frac{1}{2} < \rho \leqslant 1$, $k := 1$;

Step 1: If $\|g_k\| = 0$ then stop! else goto Step 2;

Step 2: Find a descent direction d_k that satisfies (4) and an integer $m_k > 0$;

Step 3: $x_{k+1} = x_k + \alpha_k \mathrm{d}_k$, where

$$\alpha_k = -\frac{g_k^{\mathrm{T}} d_k}{M_k \|d_k\|^2}, \tag{16}$$

in which $m_k M \geqslant M_k \geqslant \rho M$;

Step 4: $k := k + 1$, goto Step 1.

Note: In the above two algorithms, a line search procedure is avoided at each iteration. However we need to evaluate some parameters such as L_k in Algorithm (A) and M_k in Algorithm (B). For evaluating L_k, we define

$$L_k = \max\left\{\frac{\|g_i - g_{i-1}\|}{\|x_i - x_{i-1}\|} \,|\, i = k, k-1, k-2, \ldots, 2\right\}, \quad \text{for} \quad k \geqslant 2,$$

or simply take

$$L_k = \frac{\|g_k - g_{k-1}\|}{\|x_k - x_{k-1}\|}. \tag{17}$$

Certainly, at the initial step we may use only one line search.

We can also find L_k in Algorithm (A) by solving the following minimization problem

$$\min \|L_k \delta_{k-1} - y_{k-1}\|,$$

where $\delta_{k-1} = x_k - x_{k-1}$, $y_{k-1} = g_k - g_{k-1}$ and $\|\cdot\|$ denotes Euclidean norm. We obtain the solution

$$L_k = \frac{y_{k-1}^{\mathrm{T}}\delta_{k-1}}{\|\delta_{k-1}\|^2}. \tag{18}$$

Similarly, L_k can be found by minimizing the problem

$$\min \|y_{k-1} - \frac{1}{L_k}\delta_k\|$$

and the solution is

$$L_k = \frac{\|y_{k-1}\|^2}{y_{k-1}^{\mathrm{T}}\delta_{k-1}}. \tag{19}$$

The last two formulae are useful because they originate from the idea of quasi-Newton condition (e.g., [12]) as well as Barzilai and Borwein's idea [2]. The advantage of Barzilai and Borwein's gradient method is without line search. Some later conclusions on Barzilai and Borwein's method are very exciting [7,14,13,3] and enable us to investigate the descent method without line search.

As to Algorithm (B), if the Hessian matrix $\nabla^2 f(x_k)$ of $f(x)$ is easy to evaluate, then we can take

$$M_k = \max\{\nabla^2 f(x_i) | i = k, k-1, k-2, \ldots, 1\} \tag{20}$$

or take

$$M_k = L_k. \tag{21}$$

We will show that (21) is reasonable.

3. Convergence analysis

Lemma 3.1 [Mean value theorem]. *Suppose that the objective function $f(x)$ is continuously differentiable on an open convex set B, then*

$$f(x_k + \alpha d_k) - f_k = \alpha \int_0^1 d_k^{\mathrm{T}} g(x_k + t\alpha d_k)\,\mathrm{d}t, \tag{22}$$

where x_k, $x_k + \alpha d_k \in B$ and $d_k \in R^n$. Further, if $f(x)$ is twice continuously differentiable on B, then

$$g(x_k + \alpha d_k) - g_k = \alpha \int_0^1 \nabla^2 f(x_k + t\alpha d_k) d_k\,\mathrm{d}t \tag{23}$$

and

$$f(x_k + \alpha d_k) - f_k = \alpha g_k^{\mathrm{T}} d_k + \alpha^2 \int_0^1 (1-t) d_k^{\mathrm{T}} \nabla^2 f(x_k + t\alpha d_k) d_k\,\mathrm{d}t. \tag{24}$$

The above results can be seen in many publications such as (e.g., [12]).

3.1. Convergence of Algorithm (A)

Theorem 3.1. *If* (H_1) *and* (H_3) *hold, Algorithm* (A) *generates an infinite sequence* $\{x_k\}$, *and*

$$\sum_{k=1}^{+\infty} \frac{1}{m_k^2} = +\infty, \tag{25}$$

then

$$\liminf_{k\to+\infty} \left(\frac{g_k^{\mathrm{T}} d_k}{\|d_k\|}\right)^2 = 0. \tag{26}$$

Proof. By Lemma 3.1 and (H_3) we have

$$\begin{aligned} f(x_k+\alpha d_k) - f_k &= \alpha \int_0^1 d_k^{\mathrm{T}} g(x_k + t\alpha d_k)\,\mathrm{d}t \\ &= \alpha g_k^{\mathrm{T}} d_k + \alpha \int_0^1 d_k^{\mathrm{T}} (g(x_k+t\alpha d_k) - g_k)\,\mathrm{d}t \\ &\leqslant \alpha g_k^{\mathrm{T}} d_k + \alpha \int_0^1 \|d_k\| \cdot \|g(x_k+t\alpha d_k) - g_k\|\,\mathrm{d}t \\ &\leqslant \alpha g_k^{\mathrm{T}} d_k + \alpha^2 L \int_0^1 t\|d_k\|^2\,\mathrm{d}t \\ &= \alpha g_k^{\mathrm{T}} d_k + \frac{1}{2}\alpha^2 L \|d_k\|^2. \end{aligned}$$

Substituting (15) into the above formula, we have

$$\begin{aligned} f(x_k+\alpha_k d_k) - f_k &\leqslant \alpha_k g_k^{\mathrm{T}} d_k + \frac{1}{2}\alpha_k^2 L\|d_k\|^2 \\ &= -\frac{1}{L_k}\cdot\frac{(g_k^{\mathrm{T}} d_k)^2}{\|d_k\|^2} + \frac{L}{2L_k^2}\cdot\frac{(g_k^{\mathrm{T}} d_k)^2}{\|d_k\|^2} \leqslant -\left(\frac{1}{L_k} - \frac{L}{2L_k^2}\right)\frac{(g_k^{\mathrm{T}} d_k)^2}{\|d_k\|^2}. \end{aligned}$$

Noting $m_k L \geqslant L_k \geqslant \rho L$ in Algorithm (A), we obtain that

$$\frac{1}{L_k} - \frac{L}{2L_k^2} = \frac{2L_k - L}{2L_k^2} \geqslant \frac{2\rho L - L}{2L_k^2} \geqslant \frac{(2\rho-1)L}{2m_k^2 L^2} = \frac{2\rho-1}{2L}\cdot\frac{1}{m_k^2} > 0.$$

Therefore, $\{f_k\}$ is a monotone decreasing sequence. (H_1) shows that $\{f_k\}$ has a bound below and then $\{f_k\}$ has a finite limit. It follows from the above inequality that

$$\sum_{k=1}^{\infty}\left(\frac{1}{L_k}-\frac{L}{2L_k^2}\right)\frac{(g_k^{\mathrm{T}}d_k)^2}{\|d_k\|^2}<+\infty. \tag{27}$$

By (25) we have

$$\sum_{k=1}^{\infty}\left(\frac{1}{L_k}-\frac{L}{2L_k^2}\right)\geqslant\frac{2\rho-1}{2L}\sum_{k=1}^{\infty}\frac{1}{m_k^2}=+\infty.$$

The above inequality and (27) show that (26) holds. □

Corollary 3.1. *If the conditions in Theorem* 3.1 *hold and* $m_k\equiv m$ ($m>0$ *is a fixed integer*), *then*

$$\sum_{k=1}^{\infty}\frac{(g_k^{\mathrm{T}}d_k)^2}{\|d_k\|^2}<+\infty \tag{28}$$

and thus

$$\lim_{k\to\infty}\frac{(g_k^{\mathrm{T}}d_k)^2}{\|d_k\|^2}=0. \tag{29}$$

3.2. Convergence of Algorithm (B)

Theorem 3.2. *If* (H_1) *and* (H_4) *hold, Algorithm* (*B*) *generates an infinite sequence* $\{x_k\}$, *and* (25) *holds, then* (26) *holds.*

Proof. By Lemma 3.1 and (H_4) we have

$$\begin{aligned}
f(x_k+\alpha d_k)-f_k&=\alpha g_k^{\mathrm{T}}d_k+\alpha^2\int_0^1(1-t)d_k^{\mathrm{T}}\nabla^2 f(x_k+t\alpha d_k)d_k\,\mathrm{d}t\\
&\leqslant\alpha g_k^{\mathrm{T}}d_k+\alpha^2\int_0^1(1-t)\|\nabla^2 f(x_k+t\alpha d_k)\|\cdot\|d_k\|^2\,\mathrm{d}t\\
&\leqslant\alpha g_k^{\mathrm{T}}d_k+\alpha^2M\|d_k\|^2\int_0^1(1-t)\,\mathrm{d}t\\
&=\alpha g_k^{\mathrm{T}}d_k+\frac{1}{2}\alpha^2M\|d_k\|^2.
\end{aligned}$$

The remaining proof is similar to that of Theorem 3.1, and is omitted here. The proof is end. □

Corollary 3.2. If the conditions in Theorem 3.2 hold and $m_k\equiv m$ ($m>0$ is a fixed integer), then (28) holds and thus (29) holds.

3.3. Comparison with other step-sizes

In line search methods, if exact line search rule (a) or line search rule (b) is used, then we have the following relation

$$g_{k+1}^{\mathrm{T}} d_k = 0. \tag{30}$$

We denote the exact step-size by α_k^*, and the step-size used in Algorithms (A) and (B) by α_k. The following conclusions hold.

Theorem 3.3. *If* (H_1), *and* (H_3) *or* (H_4) *hold, the exact step-size* α_k^* *exists* (*including exact line search rule* (*a*) *and* (*b*)), *and* $L_k = L$ *in Algorithm* (*A*) *and* $M_k = M$ *in Algorithm* (*B*), *then*

$$\alpha_k^* \geqslant \alpha_k. \tag{31}$$

Proof. For Algorithm (A), by Lemma 3.1, (30), and Cauchy–Schwartz inequality, we have

$$\alpha_k^* L \|d_k\|^2 \geqslant \|g_{k+1} - g_k\| \cdot \|d_k\| \geqslant (g_{k+1} - g_k)^{\mathrm{T}} d_k = -g_k^{\mathrm{T}} d_k$$

and thus

$$\alpha_k^* \geqslant -\frac{g_k^{\mathrm{T}} d_k}{L\|d_k\|^2} = \alpha_k.$$

For Algorithm (B), By Lemma 3.1, (30), and Cauchy–Schwartz inequality, we have

$$-g_k^{\mathrm{T}} d_k = (g_{k+1} - g_k)^{\mathrm{T}} d_k = \alpha_k^* \int_0^1 d_k^{\mathrm{T}} \nabla^2 f(x_k + t\alpha_k^* d_k) d_k \, \mathrm{d}t \leqslant \alpha_k^* M \|d_k\|^2$$

and thus

$$\alpha_k^* \geqslant -\frac{g_k^{\mathrm{T}} d_k}{L\|d_k\|^2} = \alpha_k. \quad \square$$

Theorem 3.4. *If* (H_1), *and* (H_3) *or* (H_4) *hold,* $L_k = L$ *in Algorithm* (*A*) *and* $M_k = M$ *in Algorithm* (*B*), α_k^* *is defined by line search rule* (*c*) *or* (*d*), *then*

$$\alpha_k^* \leqslant \alpha_k, \tag{32}$$

where α_k *is yet the step-size in Algorithm* (*A*) *or Algorithm* (*B*).

Theorem 3.5. *If* (H_1), *and* (H_3) *or* (H_4) *hold,* $L_k = L$ *in Algorithm* (*A*) *and* $M_k = M$ *in Algorithm* (*B*), α_k^* *is defined by line search rule* (*e*), *then*

$$\alpha_k^* \geqslant \sigma\alpha_k. \tag{33}$$

Theorem 3.6. *If* (H_1), *and* (H_3) *or* (H_4) *hold,* $L_k = L$ *in Algorithm* (A) *and* $M_k = M$ *in Algorithm* (B), α_k^* *is defined by line search rule* (f) *or* (g), *then*

$$\alpha_k^* \geqslant (1-\beta)\alpha_k. \tag{34}$$

Note: For proofs of Theorems 3.4–3.6, refer to Shi's paper [16].

4. Convergence rate

In order to analyze the convergence rate, we restrict our discussion to the case of uniformly convex objective functions. We assume that

(H_5): f is twice continuously differentiable and uniformly convex on R^n.

Lemma 4.1. *It is assumed that* (H_5) *holds, then* $(H_3),(H_4),(H_1)$ *hold automatically and thus* (H_2) *holds,* $f(x)$ *has a unique minimizer* x^*, *and there exist* $0 < m' \leqslant M'$ *such that*

$$m'\|y\|^2 \leqslant y^{\mathrm{T}}\nabla^2 f(x)y \leqslant M'\|y\|^2, \quad \forall x, y \in R^n, \tag{35}$$

$$\frac{1}{2}m'\|x-x^*\|^2 \leqslant f(x)-f(x^*) \leqslant \frac{1}{2}M'\|x-x^*\|^2, \quad \forall x \in R^n, \tag{36}$$

$$M'\|x-y\|^2 \geqslant (g(x)-g(y))^{\mathrm{T}}(x-y) \geqslant m'\|x-y\|^2, \quad \forall x, y \in R^n \tag{37}$$

and thus

$$M'\|x-x^*\|^2 \geqslant g(x)^{\mathrm{T}}(x-x^*) \geqslant m'\|x-x^*\|^2, \quad \forall x \in R^n. \tag{38}$$

By (38) and (37) we can also obtain from the Cauchy–Schwartz inequality that

$$M'\|x-x^*\| \geqslant \|g(x)\| \geqslant m'\|x-x^*\|, \quad \forall x \in R^n, \tag{39}$$

and

$$\|g(x)-g(y)\| \leqslant M'\|x-y\|, \quad \forall x, y \in R^n. \tag{40}$$

Its proof can be found from (e.g.,[1]).

Lemma 4.2. *If* (H_1) *and* (H_3) *or* (H_4) *hold, Algorithm* (A) *with* $m_k \equiv m$ *or Algorithm* (B) *with* $m_k \equiv m$ *generates an infinite sequence* $\{x_k\}$, *then there exists* $\eta > 0$ *such that*

$$f_k - f_{k+1} \geqslant \eta\left(\frac{g_k^{\mathrm{T}}d_k}{\|d_k\|}\right)^2, \quad \forall k. \tag{41}$$

Proof. Without loss of generality, we assume that $L \equiv M$. By the proof of Theorems 3.1 and 3.2, we have

$$f_k - f_{k+1} \geqslant \left(\frac{1}{L_k} - \frac{L}{2L_k^2}\right)\left(\frac{g_k^{\mathrm{T}} d_k}{\|d_k\|}\right)^2 \geqslant \frac{2\sigma - 1}{2Lm^2}\left(\frac{g_k^{\mathrm{T}} d_k}{\|d_k\|}\right)^2.$$

By letting

$$0 < \eta \leqslant \frac{2\sigma - 1}{2Lm^2}. \quad \square$$

Corollary 4.1. *If the conditions in Lemma* 4.2 *hold, the descent direction* d_k *satisfies the angle property* (6) *at each iteration, then*

$$f_k - f_{k+1} \geqslant \eta\eta_0^2\|g_k\|^2, \quad \forall k \tag{42}$$

and thus

$$\lim_{k\to\infty} \|g_k\| = 0. \tag{43}$$

Theorem 4.1. *If the assumption* (H_5) *holds, Algorithm* (A) *with* $m_k \equiv m$ *or Algorithm* (B) *with* $m_k \equiv m$ *generates an infinite sequence* $\{x_k\}$, *and the descent direction* d_k *satisfies the angle property* (6) *at each iteration, then* $\{x_k\}$ *converges to the unique minimizer* x^* *of* $f(x)$ *at least linearly.*

Proof. We also assume that $L \equiv M$. By Corollary 4.1 and (39) we have $x_k \to x^* (k \to \infty)$.

By (42) and Lemma 4.1 we obtain

$$f_k - f_{k+1} \geqslant \eta\eta_0^2\|g_k\|^2 \geqslant \eta\eta_0^2 m'^2\|x_k - x^*\|^2 \geqslant \frac{2\eta\eta_0^2 m'^2}{M'}(f_k - f^*).$$

By setting

$$\theta = m'\sqrt{\frac{2\eta\eta_0^2}{M'}},$$

we can prove that $\theta < 1$. In fact, since $0 < \eta_0 \leqslant 1$ and $m' \leqslant L \leqslant M'$, by the definition of η in the proof of Lemma 4.2, we obtain

$$\begin{aligned}\theta^2 &= \frac{2m'^2\eta\eta_0^2}{M'} \leqslant \frac{2m'^2\eta_0^2(2\sigma - 1)}{2m^2LM'} \leqslant \frac{m'^2\eta_0^2(2\sigma - 1)}{m^2L^2} \\ &\leqslant \frac{2\sigma - 1}{m^2} \cdot \frac{m'^2}{M'^2} \leqslant \frac{2\sigma - 1}{m^2} < 1.\end{aligned}$$

By setting

$$\omega = \sqrt{1 - \theta^2},$$

which obviously means $\omega < 1$, we obtain from the above inequality that

$$f_{k+1} - f^* \leqslant (1-\theta^2)(f_k - f^*) = \omega^2(f_k - f^*) \leqslant \cdots \leqslant \omega^{2k}(f_1 - f^*).$$

By Lemma 4.1 and the above inequality we have

$$\|x_{k+1} - x^*\|^2 \leqslant \frac{2}{m'}(f_{k+1} - f^*) \leqslant \omega^{2k}\frac{2(f_1 - f^*)}{m'}$$

and thus

$$\|x_{k+1} - x^*\| \leqslant \omega^k\sqrt{\frac{2(f_1 - f^*)}{m'}},$$

which shows that $\{x_k\}$ converges to x^* linearly. □

5. Evaluation of step-size

In Algorithms (A) and (B), it is required to evaluate the step-size

$$\alpha_k = -\frac{g_k^{\mathrm{T}} d_k}{L_k\|d_k\|^2}, \quad or \quad \alpha_k = -\frac{g_k^{\mathrm{T}} d_k}{M_k\|d_k\|^2},$$

and to evaluate L_k or M_k at each iteration.

For simplicity we evaluate L_k in Algorithm (A), in which L_k satisfies

$$m_k L \geqslant L_k \geqslant \rho L,$$

where $m_k > 0$ is an integer and $0.5 < \rho \leqslant 1$.

This motivates us to estimate L_k in a wide scope. As (17)–(19) mentioned in Section 2, we have the following analysis.

For (18) with the Assumption (H_2), by the Cauchy–Schwartz inequality, we obtain that

$$L_k = \frac{y_{k-1}^{\mathrm{T}}\delta_{k-1}}{\|\delta_{k-1}\|^2} \leqslant \frac{\|y_{k-1}\| \cdot \|\delta_{k-1}\|}{\|\delta_{k-1}\|^2} = \frac{\|y_{k-1}\|}{\|\delta_{k-1}\|} \leqslant L.$$

For (19), by the Cauchy–Schwartz inequality again, we have

$$L_k = \frac{\|y_{k-1}\|^2}{y_{k-1}^{\mathrm{T}}\delta_{k-1}} \geqslant \frac{\|y_{k-1}\|^2}{\|\delta_{k-1}\| \cdot \|y_{k-1}\|} = \frac{\|y_{k-1}\|}{\|\delta_{k-1}\|}.$$

Thus it seems that (17) is more reasonable than others.

Certainly, we can take

$$L_k = \max\left\{\frac{y_{i-1}^{\mathrm{T}}\delta_{i-1}}{\|\delta_{i-1}\|^2} | i = k, k-1, \ldots, 2\right\} \tag{44}$$

corresponding to (18) and take

$$L_k = \max\left\{\frac{\|y_{i-1}\|^2}{y_{i-1}^{\mathrm{T}}\delta_{i-1}} | i = k, k-1, \ldots, 2\right\} \tag{45}$$

corresponding to (19). Taking

$$L_k = \max\left\{\frac{\|y_{i-1}\|}{\|\delta_{i-1}\|} | i = k, k-1, \ldots, 2\right\} \tag{46}$$

seems more reasonable.

6. Conclusions

In order to reduce the evaluations of objective functions and gradients at each iteration, so as to save the amount of computation, we should avoid a line search procedure in designing algorithms. In this paper we proposed a new descent method without line search for unconstrained optimization problems. The algorithm is required to estimate some parameters at each iteration. We analyzed theoretically the global convergence and convergence rate of the new algorithm under mild conditions. Further we proposed some techniques for estimating parameters used in the new algorithm. These theoretical conclusions can help us in designing new efficient methods for optimization problems. Moreover, the paper also explains that the search direction at each iteration plays a main role in descent methods. For further research we should not only find more techniques of estimating parameters but also carry out numerical experiments.

Acknowledgments

The authors are grateful to Prof. Yuan Ya-Xiang for his valuable suggestions and discussions. The work is supported in part by NSF of China (10171054), Postdoctoral Fund of China and Kuan-Cheng Wang Postdoctoral Fund of CAS (grant: 6765700).

References

[1] A.I. Cohen, Stepsize analysis for descent methods, J. Optim. Theor. Appl. 33 (2) (1981) 187–205.
[2] J. Barzilai, J.M. Borwein, Two-point step size gradient methods, IMA J. Numer. Anal. 8 (1988) 141–148.
[3] Y.H. Dai, L.Z. Liao, R-linear convergence of the Barzilai and Borwein gradient method, IMA J. Numer. Anal. 22 (2002) 1–10.

[4] Y.H. Dai, Y. Yuan, Convergence properties of Beale–Powell restart algorithm, Sci. China 41 (11) (1998) 1142–1150.
[5] X.D. Chen, J. Sun, Global convergence of a two-parameter family of conjugate gradient methods without line search, J. Computat. Appl. Math. 146 (2002) 37–45.
[6] L.C.W. Dixon, Conjugate directions without line searches, J. Inst. Math. Appl. 11 (1973) 317–328.
[7] R. Fletcher, The Barzilai–Borwein method-steepest descent method resurgent? Report in the International Workshop on "Optimization and Control with Applications", Erice, Italy, July 9–17, 2001.
[8] J.C. Gilbert, J. Nocedal, Global convergence properties of conjugate gradient methods for optimization, SIAM J. Optim. 2 (1) (1992) 21–42.
[9] L. Grippo, S. Lucidi, A globally convergent version of the Polak–Ribière conjugate gradient method, Math. Prog. 78 (1997) 375–391.
[10] L. Grippo, F. Lampariello, S. Lucidi, A class of nonmonotone stability methods in unconstrained optimization, Numer. Math. 62 (1991) 779–805.
[11] H.Y. Huang, J.P. Chambliss, Quadratically convergent algorithms and one-dimensional search schemes, J. Optim. Theor. Appl. 11 (2) (1973) 175–188.
[12] J. Nocedal, J.W. Stephen, Numer. Optim., Springer-Verlag, New York, Inc., 1999.
[13] M. Raydan, The Barzilai and Borwein method for the large scale unconstrained minimization problem, SIAM J. Optim. 7 (1997) 26–33.
[14] M. Raydan, On the Barzilai–Borwein gradient choice of steplength for the gradient method, IMA J. Numer. Anal. 13 (1993) 321–326.
[15] J. Sun, J.P. Zhang, Global convergence of conjugate gradient methods without line search, Ann. Operat. Res. 103 (2001) 161–173.
[16] Z.J. Shi, Convergence of line search methods for unconstrained optimization, Appl. Math. Computat., in press.
[17] Z.J. Shi, Restricted PR conjugate gradient method and its global convergence, Adv. Math. 31 (1) (2002) 47–55.
[18] M.N. Vrahatis, G.S. Androulakis, J.N. Lambrinos, G.D. Magoulas, A class of gradient unconstrained minimization algorithms with adaptive stepsize, J. Comput. Appl. Math. 114 (2000) 367–386.